

From Data Gloves to Deep Networks: A Survey of Sign Language Recognition and Multimodal Approach

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Abstract. More than a million deaf and hard-of-hearing people worldwide communicate through sign language. However, there is a substantial communication barrier between the deaf and hearing communities, as most hearing-abled people lack proficiency in sign language. Sign Language Recognition (SLR) systems have therefore emerged as a vital technological solution to overcome this communication divide by automatically translating sign language into spoken or written forms. This paper reviews the evolution of SLR systems, from early sensor and handcrafted-feature-based approaches to modern vision and deep-learning-based methods. We examine the progressive improvement in SLR from Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), transformer models, and multimodal architectures from isolated word-level to continuous sentence-level recognition. We also discuss the emergence of recent multimodal, large-scale datasets that promise to support robust training and evaluation of flexible, generalized SLR systems. Finally, we highlight key challenges (e.g., data scarcity, signer variability, integration of non-manual cues, and real-time performance), thereby proposing future research directions to develop universal, all-inclusive sign language recognition systems that bridge communication gaps between the deaf and hearing worlds.

Keywords: Sign Language Recognition · Deep Learning · Multimodal datasets · Vision-based Recognition · Sensor-based Recognition · Accessibility

1 Introduction

As per statistics from the Federation of the Deaf, there are about 430 million deaf and hard-of-hearing persons around the world, including more than 70 million who use the sign language (WFD, 2024). These statistics underscore the importance of sign language as the primary means of communication for people who are deaf or hard of hearing.

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Sign languages are fully developed natural languages with their own grammatical rules and structures, expressed through hand movements, facial expressions and peculiar body postures (Camgoz et al., 2018). However, they are largely unintelligible to people without prior training, creating a significant communication barrier between deaf and hearing communities. Sign Language Recognition (SLR) has therefore become an increasingly active research direction in the fields of computer vision and human-computer interaction (Camgoz et al., 2018). SLR helps to bridge this communication gap by translating sign language into spoken or written language, enhancing accessibility and inclusion in education, employment, and social life. Closely related to SLR, Sign Language Translation (SLT) extends this task by generating natural language sentences, reflecting a more comprehensive understanding of sign language (Camgoz et al., 2018).

Over the past few decades, many researchers have investigated different approaches to SLR. Table 1 illustrates the historical development of SLR.

Table 1. Evolution of Sign Language Recognition Methods

Class.	Sensor-based	Vision-based	Deep learning	Recent research
Time	1980s	1990s	2010s	2020s
Tech.	Handcrafted feat.	Image processing	CNNs	Self-supervision
	Sensors	HMM/CRF	RNNs	Transfer learning
	Motion trackers		3D CNNs	Multimodal
			Transformer	Cross-language
Feat.	High accuracy	Simpler setup	Automatic features	Better generalization
	Expensive	Lighting sensitive	Temporal modeling	Unlabelled data
	Intrusive	Limited features	Data intensive	More robust

Early related work dates back to the 1980s, when Zimmerman et al. (Zimmerman et al. 1986) developed gloves to encode the manual alphabet. This technology was applied in many early SLR studies, which relied on handcrafted features and specialized sensors such as data gloves (Mehdi and Khan, 2002). These systems captured precise motion data but were expensive, intrusive and not suitable for daily use. The accuracy achieved was more than 95%, but the results were mostly tested on small datasets, which lack practicality (Fang and Gao, 2002).

Since the 1990s, image-based approaches, particularly those using color gloves, have emerged and evolved in parallel with sensor-based models. These gloves replaced complex sensors by using either a single solid color (Starner and Pentland, 1995) or multiple colors assigned to different regions (B.Bauer and H.Hienz, 2000). This approach simplified the setup and improved hand-tracking stability, enabling cameras to capture hand features more effectively. These methods served as a middle ground between sensor-based methods and

purely vision-based methods (Starner et al., 1998), but were sensitive to lighting and background conditions.

In order to model the temporal nature of sign sequences, traditional models like Hidden Markov Models (HMM) (Starner and Pentland, 1995) and Conditional Random Fields (CRF) (Yang et al., 2009) were applied, but their performance was limited by manual feature design and further struggled with signer variability, fine-grained hand shape variations and non-manual cues.

Since 2012, the rise of deep learning has given a new direction to SLR. It replaced hand-crafted features with automatic feature learning, significantly improving recognition performance. Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) (Cui et al., 2017; Masood et al., 2018) enabled the automatic learning of spatiotemporal patterns from video data. Later models such as 3D CNNs (Al-Hammadi et al., 2020) and Transformer models (Cui, 2023; yin and Read, 2020) have further improved long-range sequence understanding, enabling more effective modelling of both local spatiotemporal features and long-range temporal dependencies and improving performance. Recent studies focus on two main tasks: isolated sign language recognition (ISLR) (Alyami et al., 2024), which recognizes isolated signs, and continuous sign language recognition (CSLR) (Hinrichs et al., 2023), which interprets entire sentences. ISLR is easier and more accurate because it recognizes signs individually, but it cannot handle continuous real world signing. CSLR can understand complete sentences, but it is harder because the signs run together and need to be segmented (Cui et al., 2017; yin and Read, 2020).

This paper presents Sign Language Recognition (SLR) techniques and provides an extensive review of the literature, highlighting their technological evolution and the associated challenges. Section 2 refers to the initial systems that utilized sensors, although accurate, their cost, limited usability, and reliance on handcrafted features limit their application. Section 3 focuses on vision based recognition systems. They overcame some limitations of sensor-based systems but faced challenges such as sensitivity to lighting, background, and individual signer differences. Section 4 examines deep learning frameworks and their impact on accuracy, usability, and generalization. CNNs and RNNs enable spatial-temporal feature learning; transformers and graph/pose models capture long-range dependencies and multi-modal end-to-end frameworks improve robustness and generalization. Section 5 discusses emerging trends and challenges in SLR.

2 Sensor-Based Recognition Systems

In the early stages, studies on SLR mainly relied on sensor-based systems and handcrafted features. Researchers used data gloves or motion trackers to record finger bending and hand trajectories (Mehdi and Khan, 2002; Fang and Gao, 2002). Although these systems captured precise motion data, they were expensive and intrusive because users had to wear bulky or wired gloves, calibrate them frequently, and use special hardware, making them unsuitable for daily use.

2.1 Feature-Based Recognition Methods

Early SLR systems treated the outputs of data gloves and motion trackers as manually defined feature vectors. Typical features included finger joint angles, hand position and orientation, and movement trajectories. These handcrafted features were then fed into traditional sequence models to recognize isolated (Mehdi and Khan, 2002) or continuous signs (Fang and Gao, 2002). Feature-based methods established a foundation for SLR research by demonstrating that structured, temporal modeling of hand gestures could support continuous recognition. This laid the groundwork for later vision-based approaches that aim to learn rich, discriminative features directly from video.

2.2 Classical Machine Learning Models

To classify the extracted features, traditional machine learning algorithms such as Support Vector Machines (SVM) and Random Forests were commonly employed, while temporal dependencies and sequence alignment were modeled using Conditional Random Fields (CRF) or Hidden Markov Models (HMM) (Fang and Gao, 2002). These models provided a structured way to capture temporal patterns in gestures, but their performance depended heavily on the quality of the input features and extensive parameter tuning. As a result, they struggled to generalize across different datasets, signers, and environments.

2.3 Limitations

Despite their contributions, early approaches exhibited several limitations. They were unable to effectively handle continuous signing, complex backgrounds, and inter-signer variability. They could not learn abstract feature representations automatically, limiting their scalability to large, diverse datasets. This limitation led the research to focus on vision-based recognition systems.

3 Vision-Based Recognition Systems

To overcome the limitations of sensors for signers, vision-based methods were introduced using color gloves or cameras (Starner and Pentland, 1995). The gloves use colored regions to simplify hand and finger segmentation. Unlike systems that relied on data gloves or motion sensors, vision-based recognition uses RGB camera to capture visual information about hand movements, shapes, and positions in a natural and non-intrusive manner. This section reviews representative studies that utilize RGB cameras for SLR.

3.1 Early RGB-Based Systems

These approaches relied on handcrafted visual features extracted from RGB camera feed. Starner and Pentland (Starner and Pentland, 1995) proposed one

of the earliest real-time continuous American Sign Language recognition systems. Their approach used video featuring colored gloves to facilitate hand tracking and extracted features such as hand position, orientation, and bounding ellipse properties. The system used Hidden Markov Models (HMMs) and achieved a word accuracy of 99.2% on an independent set of 99 test sentences, demonstrating that real-time continuous recognition could be achieved with relatively simple visual features. Building on this, Bauer and Hienz (B.Bauer and H.Hienz, 2000) developed one of the first CSLR systems based solely on a single color video camera and colored gloves. They extracted manual parameters, and modeled them using continuous-density HMMs, achieving 91.7% accuracy for a 97-sign vocabulary of German Sign Language, further corroborating the viability of video input for SLR. These experiments clearly demonstrated the initial success of vision-based systems for real-time recognition of continuous sign languages and paved the way for future research towards marker-less and more natural video recording methods.

3.2 Further Development of RGB-Based Systems

Subsequent studies of RGB-based systems have focused on improving both the naturalness of gestures and the usability. AL-Rousan et al. (AL-Rousan et al., 2009) developed a vision-based Arabic SLR system based upon HMM that operates without gloves, allowing signers to perform gestures freely and naturally. Their approach allows for both online and offline sign recognition without signer independence constraints, yielding high accuracy (96.74% on test data in offline signer-dependent mode, and 90.6% in online signer-independent mode) while processing real-world apparatus applications. Later Okayasu et al. (Okayasu et al., 2017) explored a lightweight and low-cost RGB-based approach using colored gloves and HMM. Their approach of SLR using HMM is depicted in Fig. 1. They reported a recognition rate of 62.3% on a 25-word data set. Although the accuracy is lower than the earlier lab-oriented systems, the study proved the viability of SLR on portable RGB platforms, and supported the continued focus on a more functional and easily accessible vision-based SLR system.

3.3 Comparison

These all were important approaches in the development of vision-based SLR systems. The initial works showed that CSLR could be carried out via handcrafted visions and HMMs, proving the feasibility of the RGB-based approaches and moving away from systems that rely exclusively on sensors. Later RGB-based systems emphasized simplicity, portability, and low computational requirements, making them more suitable for real-world applications. Even if the accuracy of the recognition systems were low, the important aspect demonstrated by the studies were the convenience of use, real-time capability, and limited hardware requirement. However, despite the progress made, the models were heavily reliant on hand-designed features. This posed significant challenges for generalization capabilities in the presence of variations due to lighting conditions and signer

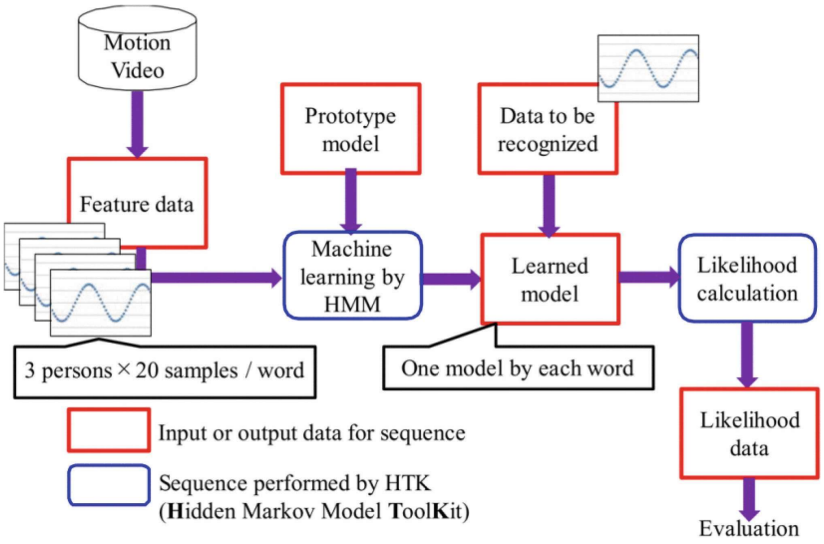


Fig. 1. SLR by HMM (Okayasu et al., 2017).

variations. These shortcomings paved the way for the adoption of deep learning models for research and the development of more discriminative models.

4 Deep Learning Techniques

Deep learning has greatly improved the accuracy rate of SLR (Cui et al., 2017). Unlike conventional algorithms that use hand-designed features or statistical models like HMM and CRF, deep learning allows hierarchical feature learning with the ability to model spatial and temporal dependencies in sign language videos. With CNNs, RNNs, and transformers, SLR algorithms have achieved tremendous success in ISLR and CSLR. Large annotated datasets have also fueled the use of deep learning, achieving the accuracy level of humans in constrained settings (Camgoz et al., 2018). This section discusses major developments in the state of the art of deep learning-based SLR, from early CNN-RNN pipeline-based systems to the latest transformer-based, graph-based, and multimodal approaches, and summarizes performance evaluation.

4.1 Convolutional and Recurrent Architectures

Convolutional and recurrent architectures have been widely applied in both ISLR and CSLR (Maruyama et al., 2024). A typical CNN-RNN pipeline is shown in Fig. 2. CNNs are utilized for video frames features extractions for hand shapes, locations, orientations, and RNNs are used to track dynamic patterns across frames. Various studies, such as Masood et al. (Masood et al., 2018), have shown that integrating convolutional layers with RNNs achieves accurate recognition of sequential patterns in gestures.

There has been substantial advancement in the field of deep learning, which has greatly improved the accuracy of SLR. Although early studies also achieved

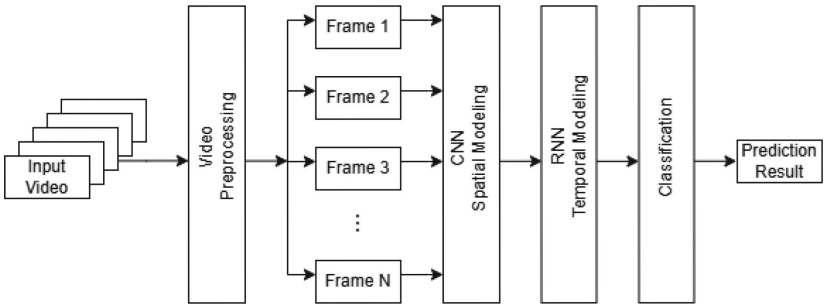


Fig. 2. Typical CNN-RNN pipeline for video-based SLR. The input video is first preprocessed into a sequence of frames, which are passed through a CNN for spatial feature extraction. The feature sequence is then fed into RNN to model temporal dependencies and predict the final gloss.

relatively high accuracy, they were conducted in controlled laboratory environments and evaluated on small-scale datasets.

Subsequent work further improved the approach using both 3D CNNs and hybrid CNN LSTM architectures (Al-Hammadi et al. 2020; Maruyama et al., 2024; Koller et al., 2020), that integrate motion as well as spatial cues directly from raw video volumes, significantly improving temporal modelling and overall accuracy. These architectures remain the foundation of modern SLR pipelines, particularly in low-resource and embedded applications.

4.2 Transformer-Based Approaches

Transformer networks have been increasingly applied in the field of SLR and have gradually replaced conventional methods. By combining CNN-based feature extraction with transformer architectures, the model outperforms traditional RNN-based methods (De Coster et al., 2020), as self-attention captures long-range dependencies more effectively.

Transformer models can also be combined with handcrafted feature extractors. De Coster et al. (De Coster et al., 2020) compared Transformer and LSTM models using the same feature extractor (OpenPose), and the results show that the Transformer outperforms the LSTM (61.73% vs 54.55% in Top-1 accuracy). However, when compared with a CNN-Transformer model (Top-1 = 73.39%), the results indicate that CNN-based methods perform better than handcrafted feature extractors. This is because CNNs can directly learn features from raw data, while OpenPose may lose important information. Despite this, handcrafted feature-based approaches still show potential for computationally efficient deployment (Boháček and Hrúz, 2022), especially on handheld or embedded devices, due to their low-dimensional representations, which reduce the complexity of downstream models (Fig. 3).

Studies also shown that pure transformer-based architectures can also be applied to video tasks (Arnab et al., 2021) as it effectively captures the dependencies along both spatio-temporal dimensions. However, these approaches typically require large datasets and high computational resources due to their parameter

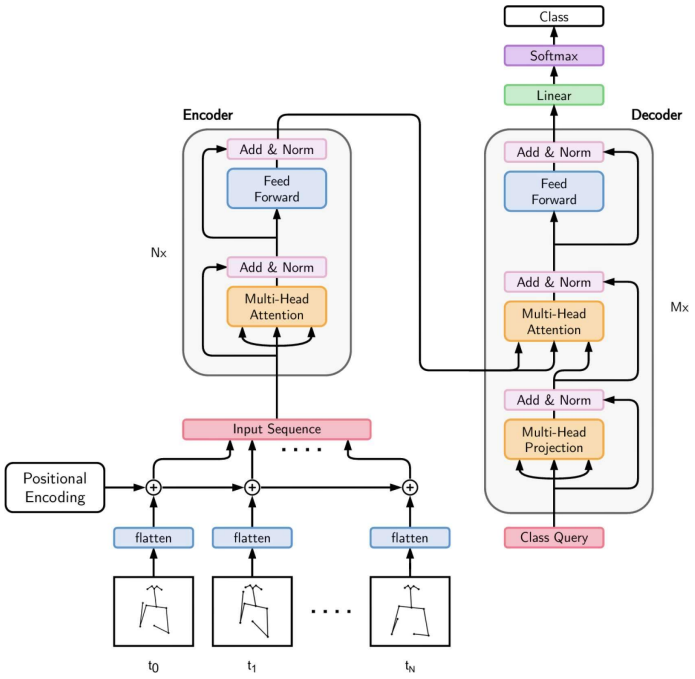


Fig. 3. A SLR system with a standard Transformer(Boháček and Hrúz, 2022).

size and the quadratic complex of global self-attention. These tend to overfit on small-scale datasets. A practical solution to this problem is to incorporate spatio-temporal modeling components within the transformer, as demonstrated by the STMC-Transformer (Yin and Read, 2020). Such modules explicitly capture sign language motion patterns, preserving fine-grained hand and body movements, which reduces the reliance on large datasets and enables strong performance on small-scale corpora.

4.3 Pose and Graph-Based Learning

Apart from pixel-based representation methods, pose and graph-based approaches have emerged as alternative for modeling human body dynamics in the SLR task (Miah et al., 2024). The development of skeleton-based action recognition has made it possible to extract human skeletal poses from regular videos (Yan et al., 2018). This has emerged as the technical foundation for pose-based SLR, where human pose estimation tools such as OpenPose or MediaPipe are employed to extract key points that define the signing individual's hands, arms and facial expressions. Building upon pose-based representations, graph-based approaches can further model relationships among keypoints by representing them as graph nodes and edges (Tunga et al., 2021). This enables the use of the Graph Neural Network (GNN) or Spatio-Temporal Graph Convolutional Network (ST-GCN) to capture spatial and temporal dependencies among the key points.

Such models will be inherently invariant to background clutter and lighting changes, and hence robust on real-world data. In fact, recent studies have

already demonstrated that fusing pose-based inputs with RGB or depth modalities improves the recognition performance even further, offering a good balance in efficiency versus semantic richness (Maruyama et al., 2024).

4.4 Multimodal and End-to-End Frameworks

The previous discussion has mainly focused on feature extraction and temporal modeling. In addition to these aspects, two other core trends are multimodal learning and end-to-end modeling (Camgoz et al., 2018), which improve model robustness and generalization (Camgoz et al., 2020; Lin et al., 2023). By incorporating several input modalities during training, such as image inputs, pose keypoints, skeletal information, and facial information, models can capture the performed actions comprehensively (Maruyama et al., 2024). Multistream learning with multiple modalities of inputs has shown to substantially enhance recognition accuracy (Maruyama et al., 2024).

End-to-end modeling approaches map the raw video input to glosses and/or textual form (Lin et al., 2023). These models incorporate spatial, temporal, and contextual representations in a holistic manner, making them better to handle complex continuous data (Yin and Read, 2020). Additionally, techniques such as pose estimation, data imputation, and velocity feature augmentation, strengthen temporal representations and contribute to state-of-the-art performance (Hinrichs et al., 2023). Multimodal end-to-end models can serve as a platform for future advancements in real-time translation and cross-lingual recognition (Lin et al., 2023).

4.5 Performance Evaluation

Performance in deep learning-based SLR is typically evaluated on benchmark datasets using metrics such as accuracy and Word Error Rate (WER). Table 2 summarizes the progression of SLR techniques by presenting representative models and their corresponding performance on the RWTH-PHOENIX-Weather 2014 dataset (Forster et al., 2012).

Table 2. Comparison of methods on PHOENIX-2014 datasets (WER(%)).

Method	WER(%)	Reference
CNN+RNN	38.7	(Cui et al., 2017)
CNN+LSTM+HMM	26.8	(Koller et al., 2017)
Multimodal SignBERT	20.2	(Zhou et al., 2021b)
HST-GNN	19.5	(Kan et al., 2022)
Swin Transformer	18.7	(Alyami and Luqman, 2025)

Early CNN-based methods achieved moderate performance with WER of more than 25%. The introduction of transformer-based architectures and mul-

timodal approaches has substantially reduced error rates, demonstrating superior modeling of long-term dependencies and sequence-level information. Graph-based models further improve robustness by incorporating skeleton structure information. Overall, the Table 2 highlights the clear trend that newer deep learning architectures consistently outperform earlier CNN and RNN-based methods.

5 Emerging Trends and Challenges

In addition to the mainstream CNN/RNN and Transformer-based models, a number of advanced exploratory approaches have emerged in the field of SLR. Some models have achieved state-of-the-art performance on widely used benchmarks such as PHOENIX-2014 and CSL-Daily (Zhou et al., 2021a). For instance, Zhang et al. incorporate cross-modal contextual information by integrating a language model and introduce the Contextualized Sequence Transduction (CST) loss to effectively model dependencies between glosses, thereby demonstrating strong performance in long-sequence prediction (Zhang et al., 2023). However, it suffers from high computational cost and training complexity. Diffusion-based CSLR reformulates CSLR as a generative task (Geng et al., 2026), directly producing gloss sequences via diffusion models, thereby avoiding errors. Slow-Fast CSLR adopts a dual-pathway architecture to capture low-frame-rate spatial features and high-frame-rate dynamic features (Ahn et al., 2024), and further improves representation through bidirectional feature fusion and auxiliary enhancement modules. Guan et al. utilize keypoint sequences as input and employ a multi-stream self-attention mechanism along with self-distillation (Guan et al., 2025). It is robust to background variations, computationally efficient, and easily extendable to SLT tasks but highly depends on keypoint quality. These approaches advance SLR/SLT research from different perspectives. However, due to their general complex architectures, higher training difficulty and deployment challenges, these have not yet become mainstream methods.

Another key trend is the development of large multimodal datasets that capture multiple channels of communication, including hand motion, body posture and facial expressions (Shen et al., 2024). In parallel, recent research has moved beyond well-studied sign languages such as American Sign Language (ASL), Chinese Sign Language (CSL) and German Sign Language (GSL), toward the creation of large-scale datasets for low-resource sign languages. A representative example is Australian Sign Language (Auslan), for which new datasets such as MM-WLAuslan (Shen et al., 2024) and Auslan-Daily (Shen et al., 2023) have been introduced. These datasets provide hundreds of thousands of annotated video samples and support tasks including ISLR, CSLR and SLT. MM-WLAuslan can be used for ISLR. The key advantage of this dataset is its strong balance between scale and data quality. It is larger than early datasets such as Purdue RVL-SLLL and RWTH-BOSTON, with 3215 signs, 282.9k videos and 73 signers, which can help improve model generalization (Shen et al., 2024). Compared with large-scale web-based datasets like MS-ASL and WLASL, MM-WLAuslan is recorded in professional studios, resulting in higher video qual-

ity and lower noise. However, it may lack naturalness and real-world variability. Auslan-Daily is suitable for CSLR. It contains 25k videos, 14k vocabulary items, and 67 signers, presenting a moderate-scale dataset. Its key advantage is its inclusion of more everyday communication scenarios, making it closer to real-world usage. Additionally, it features diverse signer positions and multi-person interactions(1–10 persons per clip), which enhance model applicability and robustness in natural sign language streams (Shen et al., 2023).

Other notable trends include efforts to improve generalization across signers and environments (Hu et al., 2021, 2023; Najib, 2025). In addition, cross-lingual and multilingual models have been explored to share visual and linguistic knowledge across different sign languages, enabling better transfer between languages with similar structures (Yin et al., 2022). These approaches help move toward sign language systems that can understand and translate signs from many different communities around the world.

Despite these advances, several persistent challenges continue to limit the widespread application of SLR. Data scarcity and annotation costs remain major bottlenecks, further complicated by linguistic diversity across different sign languages. In addition, signer variability including differences in appearance, motion, and camera viewpoint limit model performance and generalization in unconstrained environments. Another challenge is modeling non-manual components such as facial expressions, gaze, and mouth movements which are essential for grammatical and emotional information, but current models still struggle to integrate them effectively. In addition, real-time inference and deployment are constrained by the high computational cost of deep models. Balancing recognition accuracy with speed and energy efficiency is also a key area of future research. Finally, ethical and social considerations such as privacy, inclusivity, and accessibility must be addressed. Overall, while recent developments in self-supervised learning, multimodal fusion and cross-lingual modelling represent exciting progress, achieving robust, inclusive, and socially responsible SLR systems remains an ongoing research challenge.

6 Conclusion

This review paper has presented the evolution of SLR from early sensor-based and handcrafted approaches to modern deep learning architectures. Early systems achieved good tracking but suffered from high cost and limited practicality. The shift toward vision-based recognition, initially relying on colour gloves and handcrafted features, laid the foundation for data-driven learning. With the rise of deep neural networks, especially CNNs, RNNs and Transformers, SLR has advanced in terms of spatiotemporal modelling and end-to-end translation capability. The development of large-scale multimodal datasets further advances research in SLR. Recent research further integrates multimodal inputs, improving robustness and semantic understanding. Despite these achievements, several challenges remain, including data scarcity, signer variability, and the difficulty of modelling non-manual features. By combining advances in deep learning with

diverse, well-annotated corpora, future research can move toward universal, real-time, and socially inclusive sign language translation systems.

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